

Visualization

An Introduction

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THE UNIVERSITY OF
CHICAGO

UIC

CITY COLLEGES
of CHICAGO

HOWARD
UNIVERSITY

NC STATE
UNIVERSITY

M
MOREHOUSE

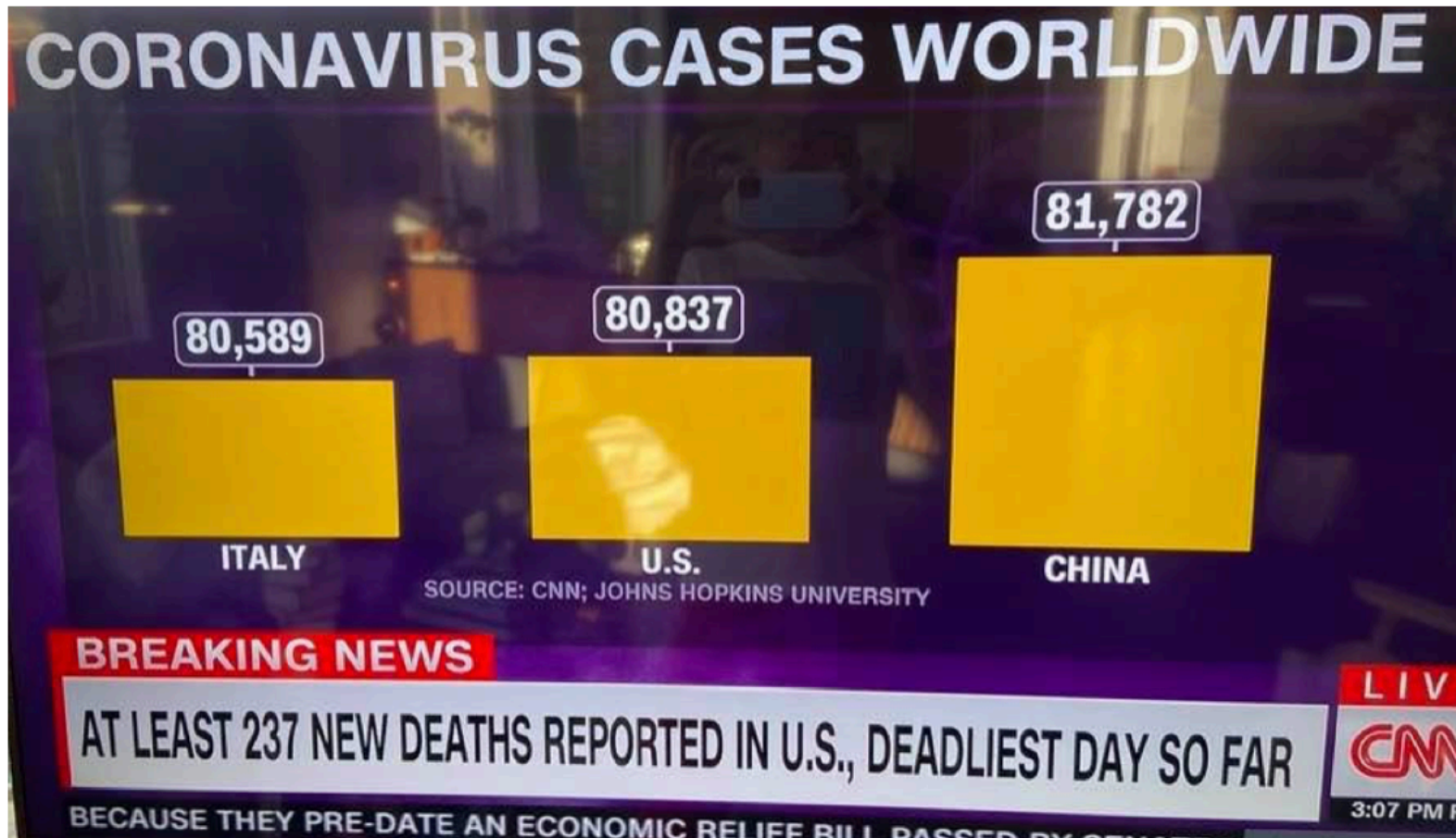
FRESNO
STATE
Discovery
Diversity
Distinction

UTSA
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at San Antonio™

Goals

- Visualization introduction
- communicating your data

Visualization...What not to do



IF BUSH TAX CUTS EXPIRE

TOP TAX RATE



8:01p ET

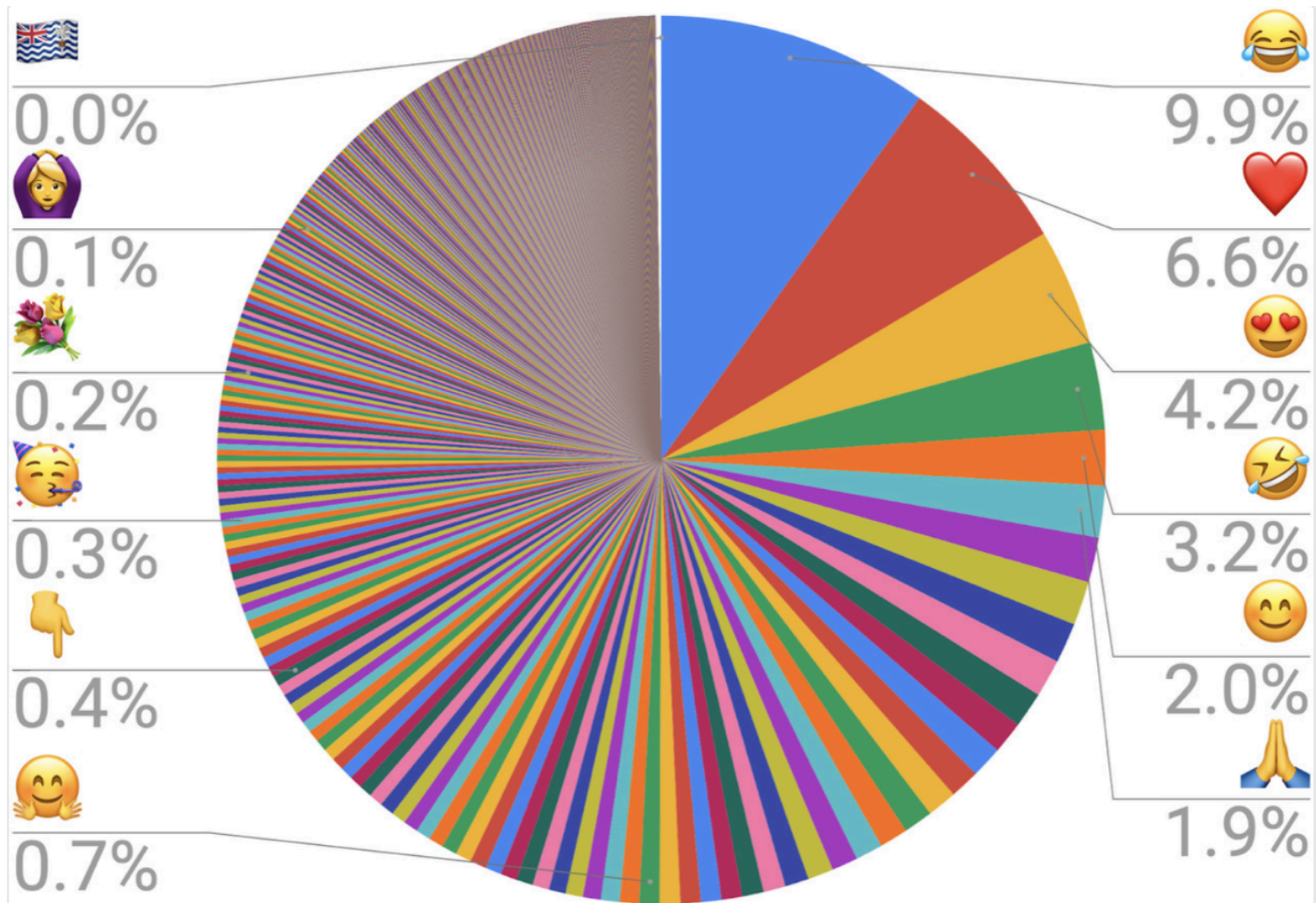
FOX
BUSINESS

TOP STORIES

TECHNOLOGY

CONSUMER

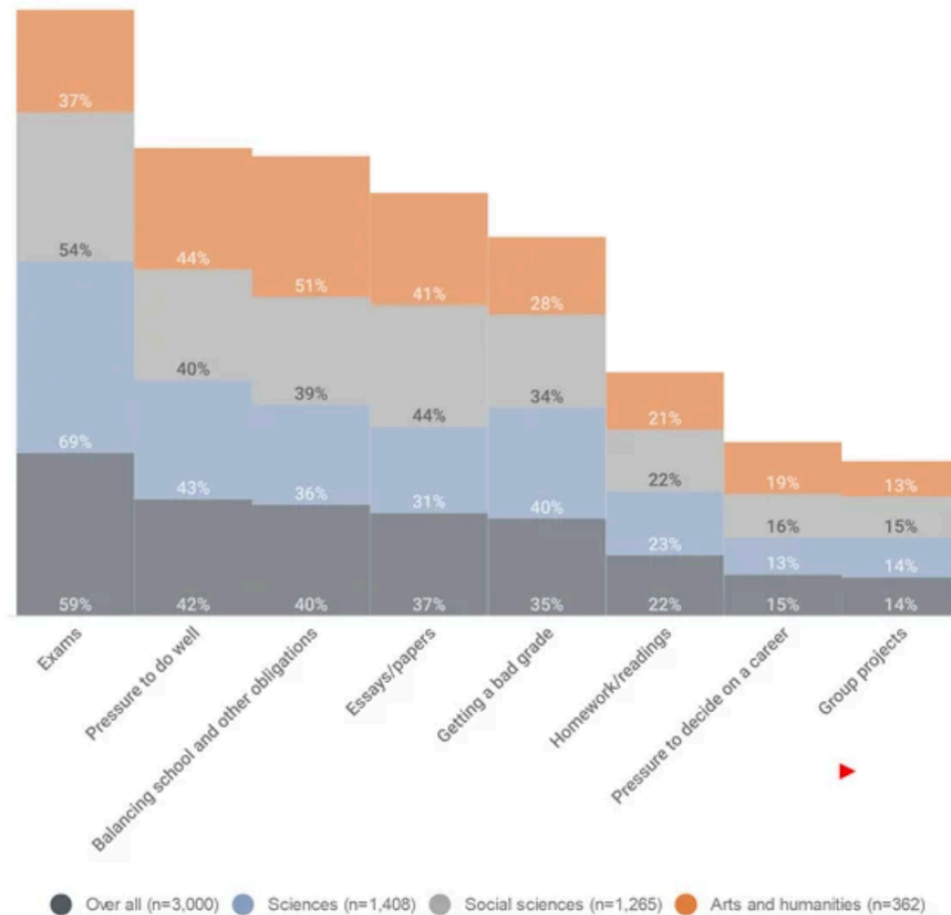
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The Most Frequent Emoji

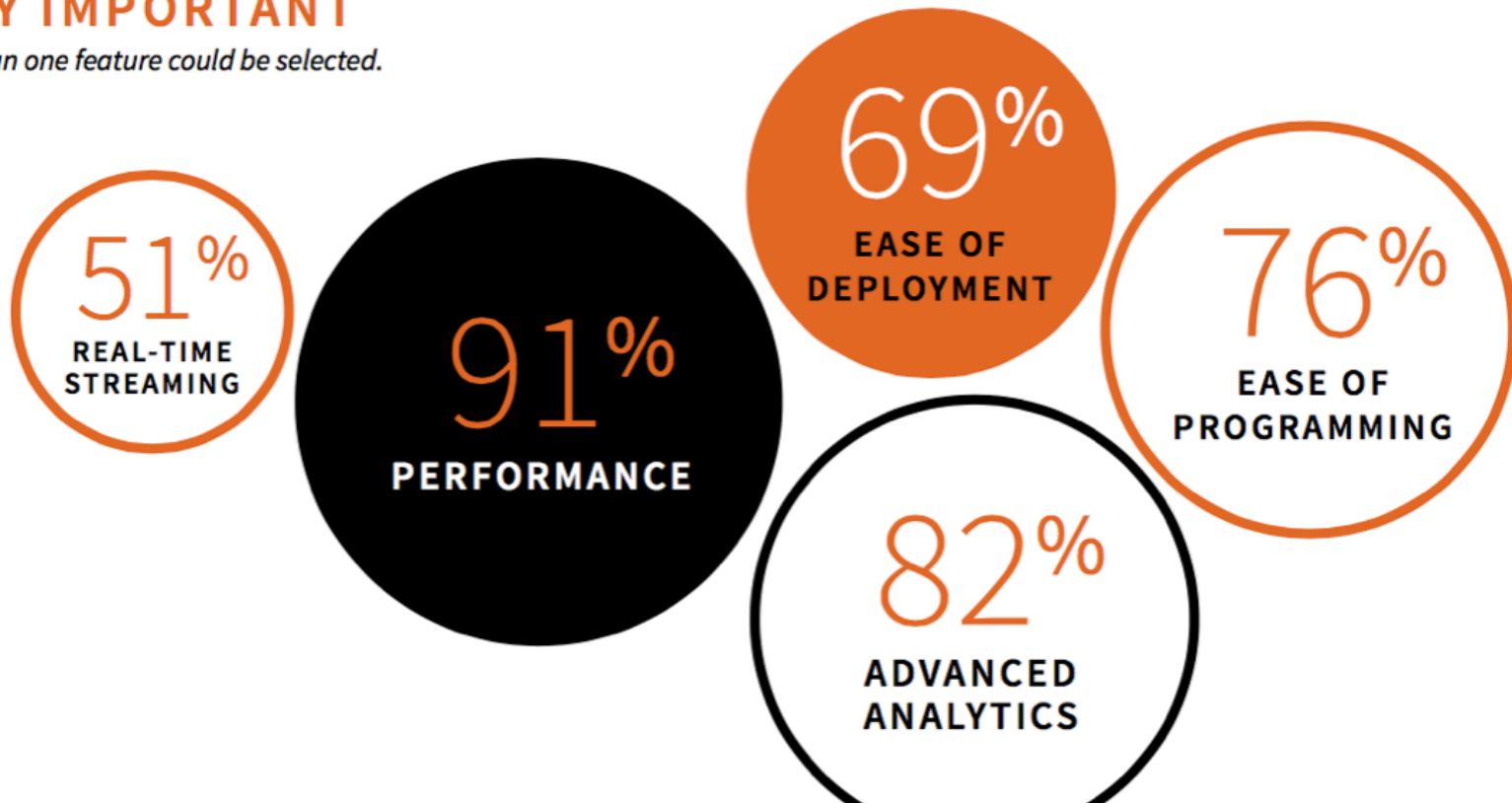
Students' Biggest Academic Stressors

Top eight contributors to students' academic stress, by share of respondents over all and by major (could select three factors from a list of 12)



% OF RESPONDENTS WHO CONSIDERED THE FEATURE VERY IMPORTANT

More than one feature could be selected.



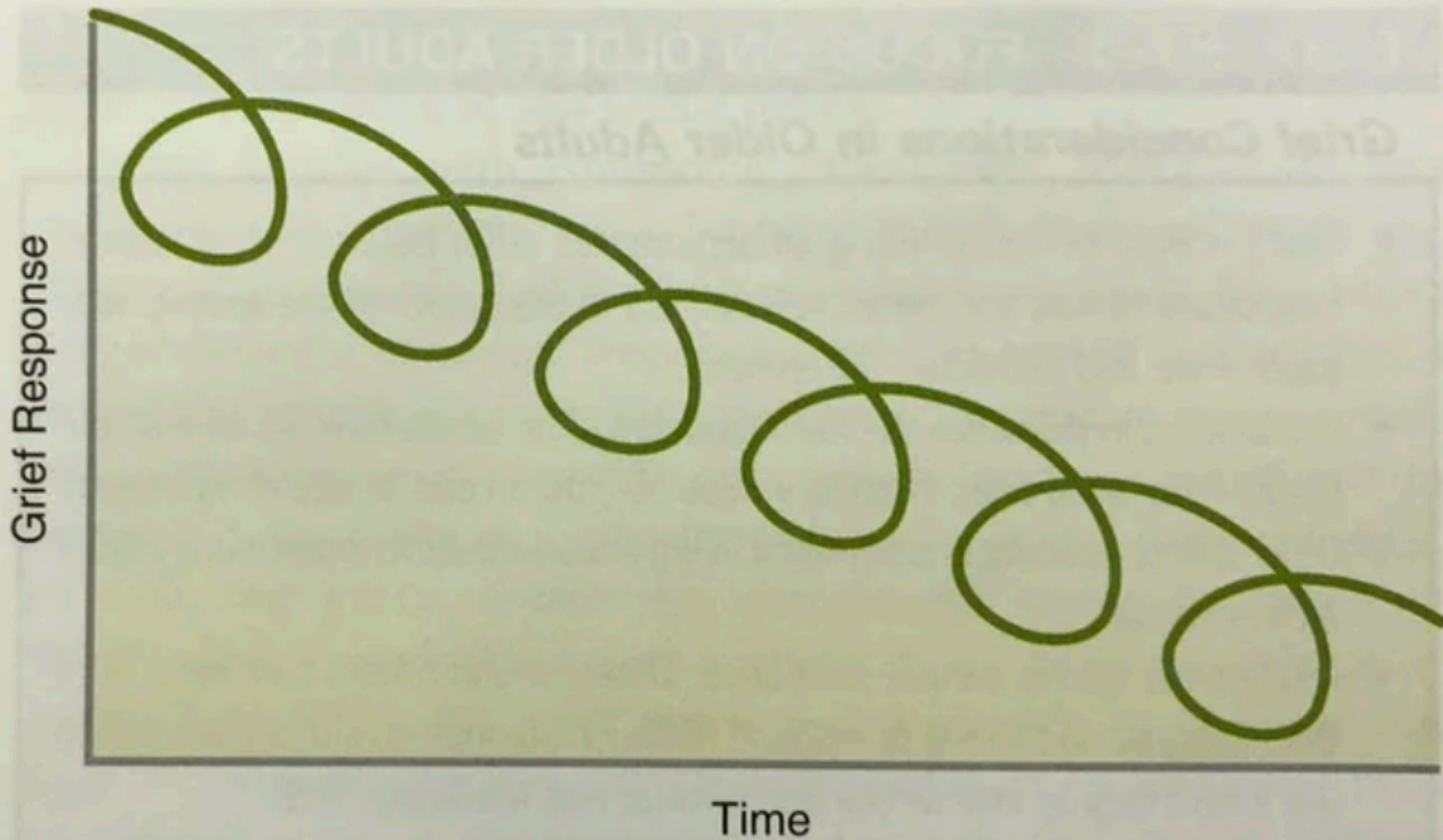
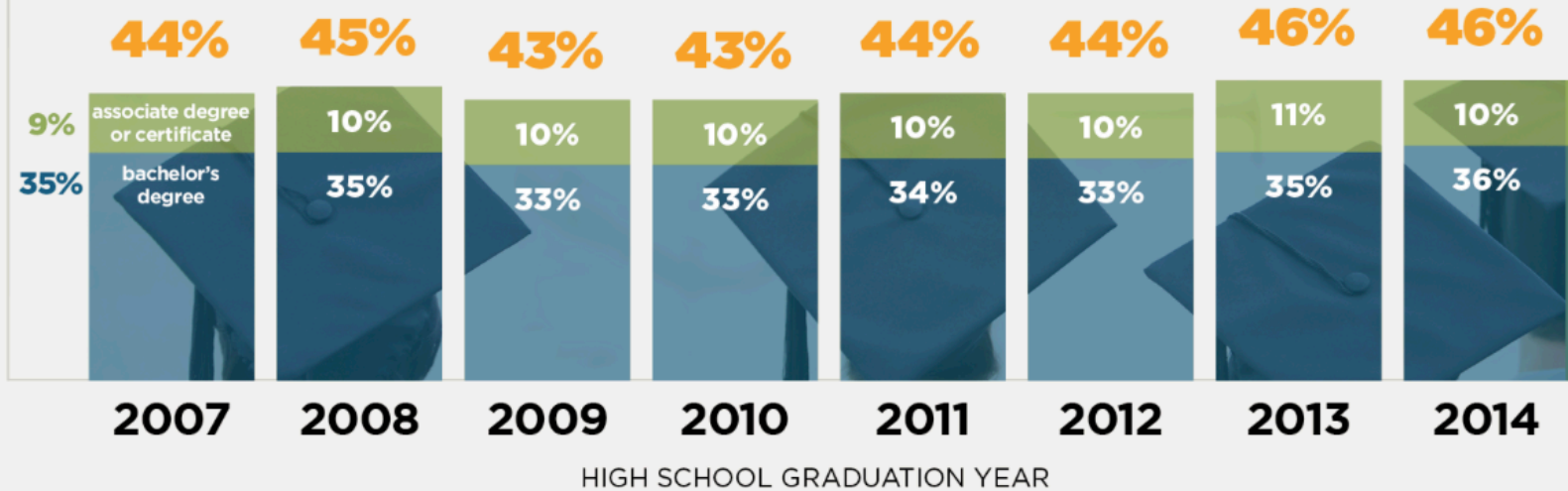


FIGURE 37-1 Cyclical nature of a normal grief response over time.

After years of holding steady, the CPS college completion rate reached 46 percent among 2013 and 2014 CPS graduates

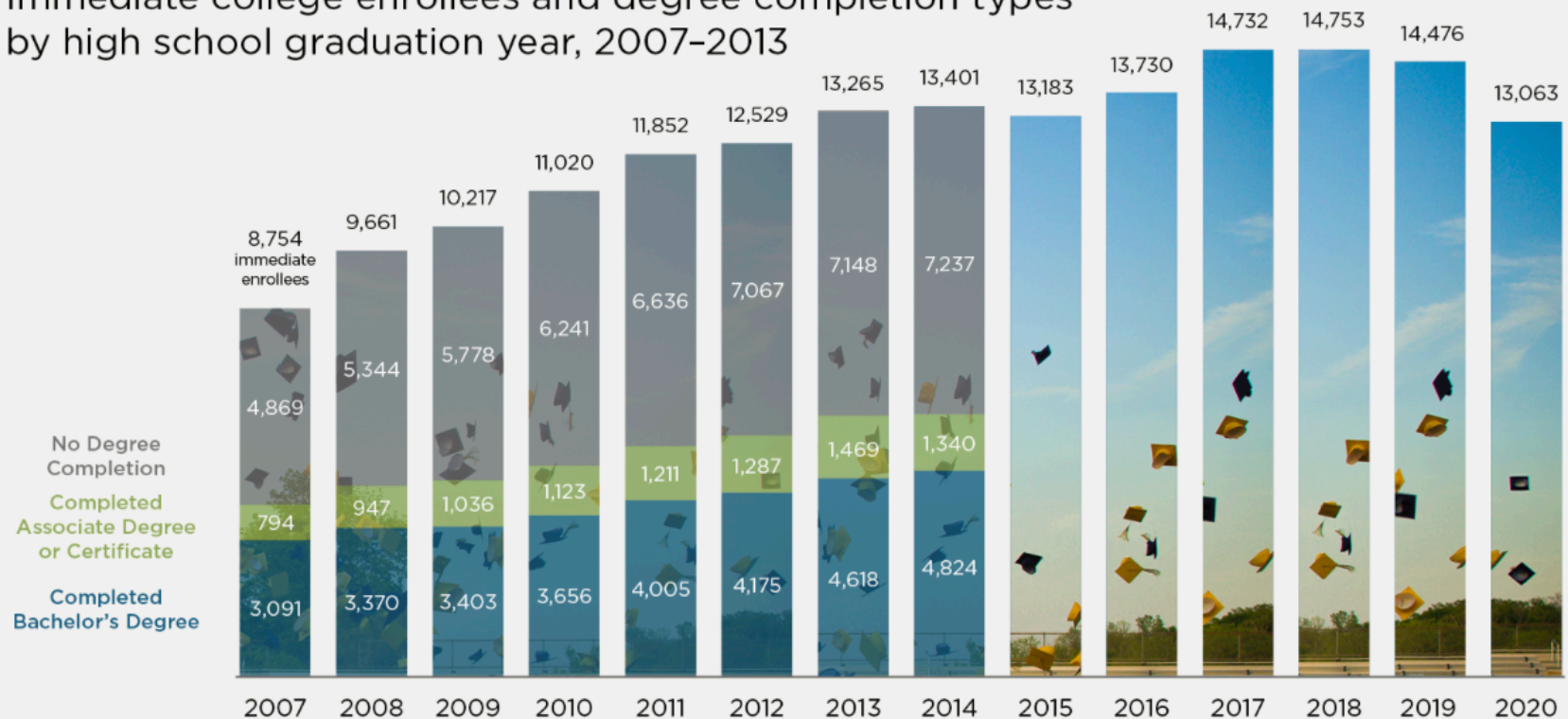
SIX-YEAR COLLEGE COMPLETION RATE

Six-year college completion among immediate college enrollees in the CPS graduating classes of 2007-2014



More students have immediately enrolled in college: hundreds more earned a degree, and hundreds more left without one

Immediate college enrollees and degree completion types by high school graduation year, 2007-2013



Important Take-aways!

- Be thoughtful about *who* your visualization is for
- Present your data in a clear way
 - title
 - labels

Data Types

Numerical — Each value is from a fixed scale - Can have *discrete* or *continuous* data

- Discrete: finite, countable - integers
 - age, number of kids
- Continuous: infinite, take on any real number in an interval
 - income, height, test scores

Numerical measurements are ordered

Differences are typically meaningful

Categorical — Each value is from a fixed inventory

- Can have *nominal* or *ordinal* data
- Nominal: a naming or labeling scale (that is NO natural order is present)
 - hair color, names of people
- Ordinal: ordered rank (natural order is present)
 - Letter grade, customer satisfaction (very satisfied...very dissatisfied), education

Categories are the same or different

Numerical Data

When working with numerical data, it is usually most appropriate to visualize through

- scatter plots
- line plots
- histograms

Categorical Data

When working with categorical data, it is usually most appropriate to visualize through

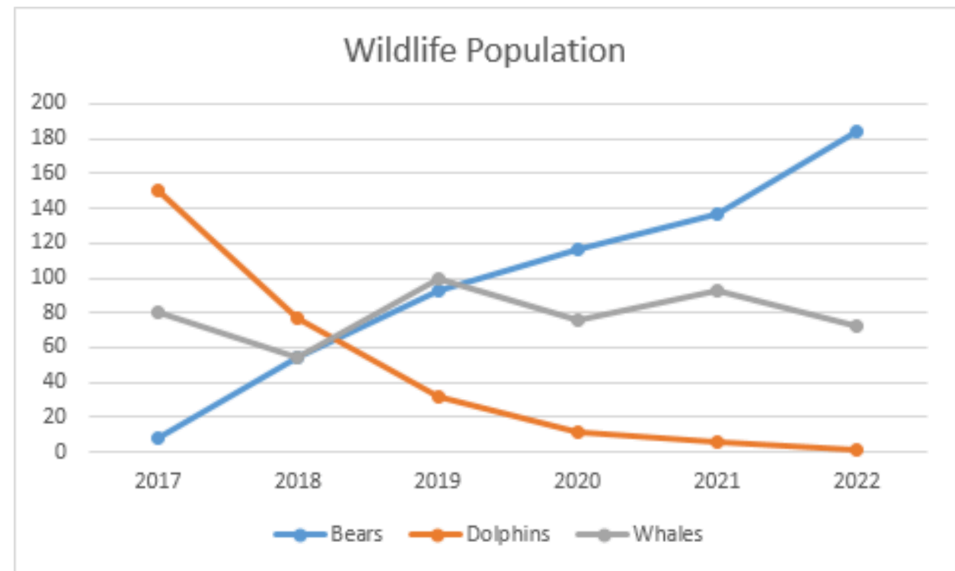
- bar plots
- pie charts

For Numerical data

- **Line plots**
 - useful when the x-axis is ordered (e.g., time data)
- **Scatter plots**
 - useful for comparing two dimensional data

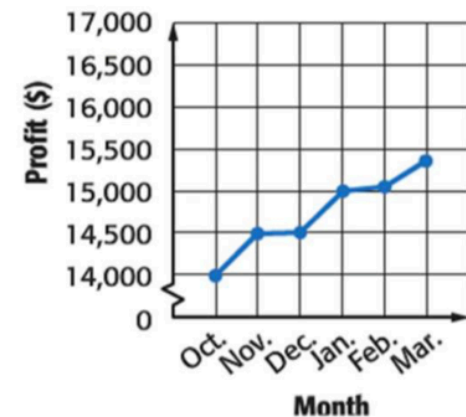
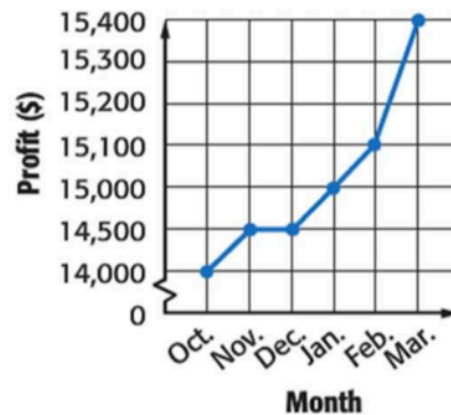
Line Graphs

- Best to show numerical values over a continuous interval or over time
- Nice way to show trends
- One output for each input



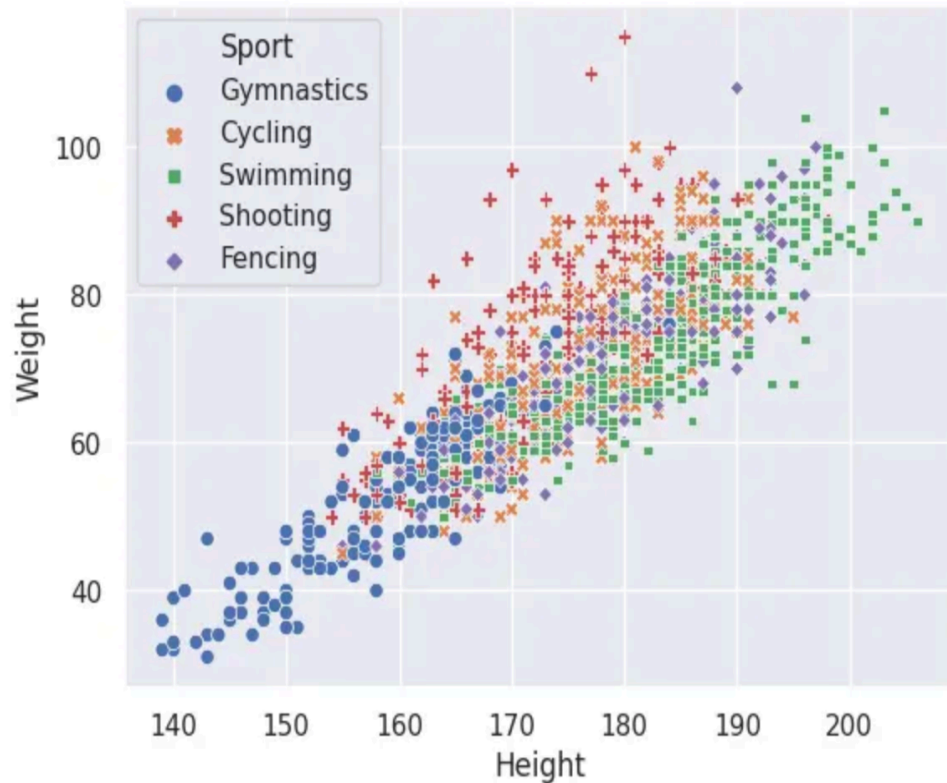
Be aware of possible issues!

- Use consistent interval length
- Use the full axis and do not skip values when working with numerical data



Scatter plots

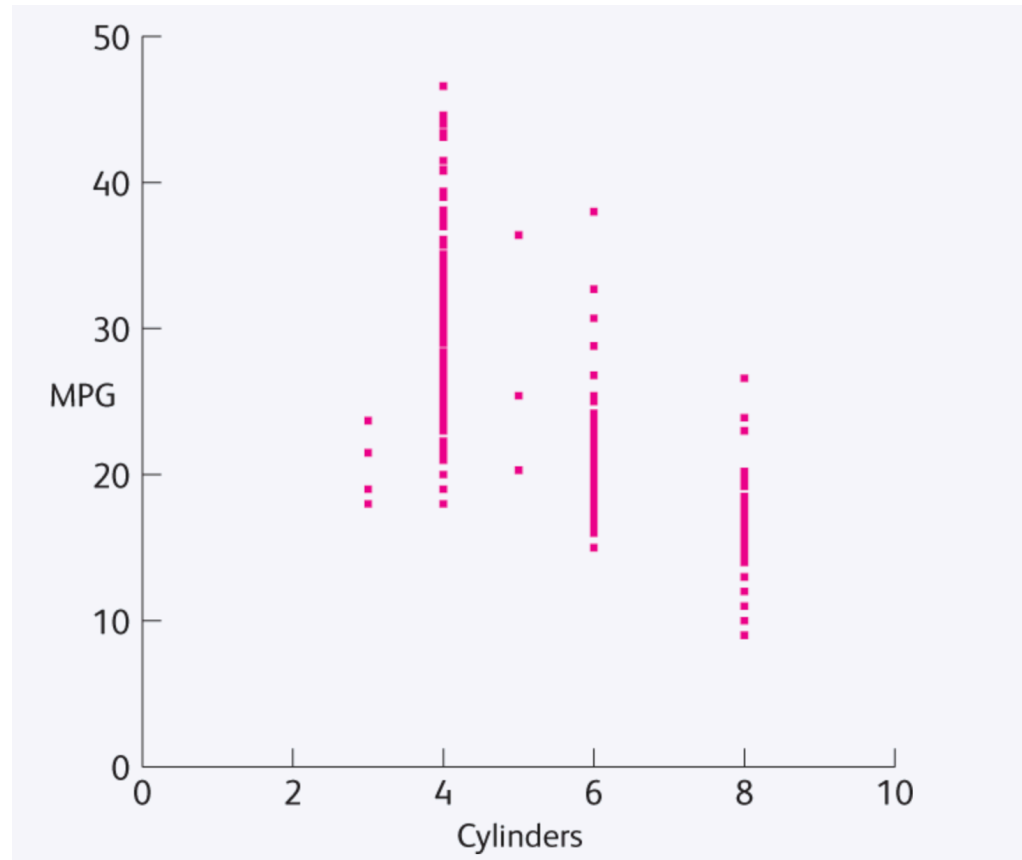
- Best to compare two numerical variables
- multiple outputs (potentially) for each input



Be aware of possible issues

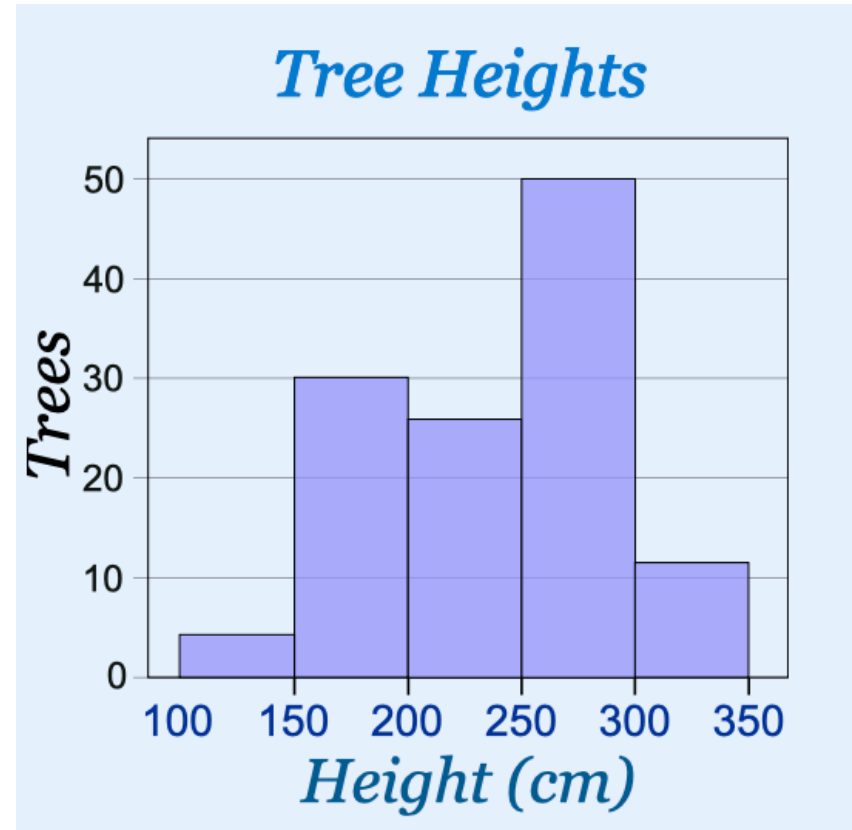
- Rounded numerical values (decimal places)
- Inaccurate or not very precise measurements

This may cause numerical data to appear categorical!



Histogram

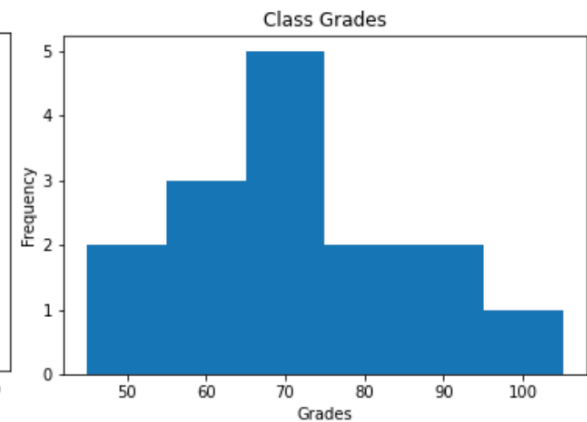
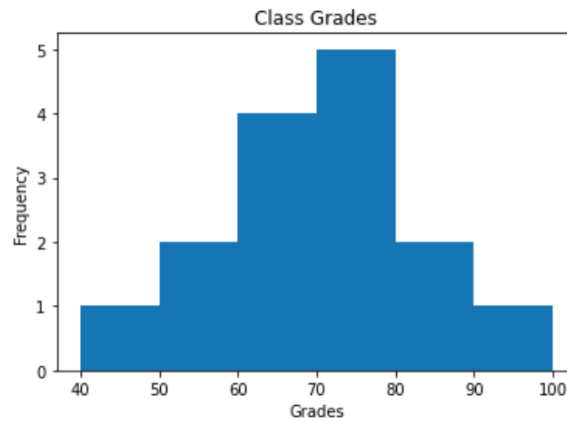
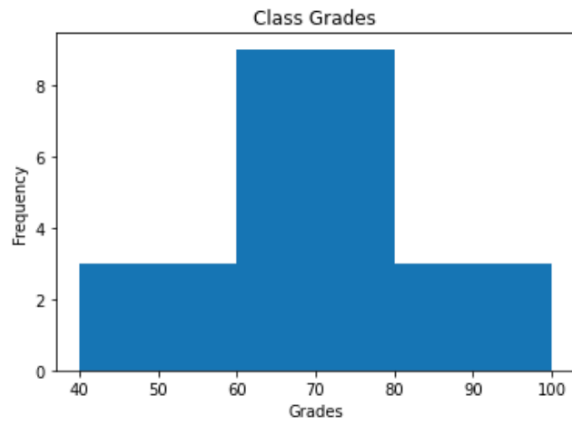
- Gather data into bins (numerical range)
- A bar in a histogram represents the frequency of that range



There are 50 trees with height from 250-300 cm.

Be aware of possible issues

Histograms are extremely sensitive to the number of bins and bin width!



For Categorical Data

Bar plots (bar charts)

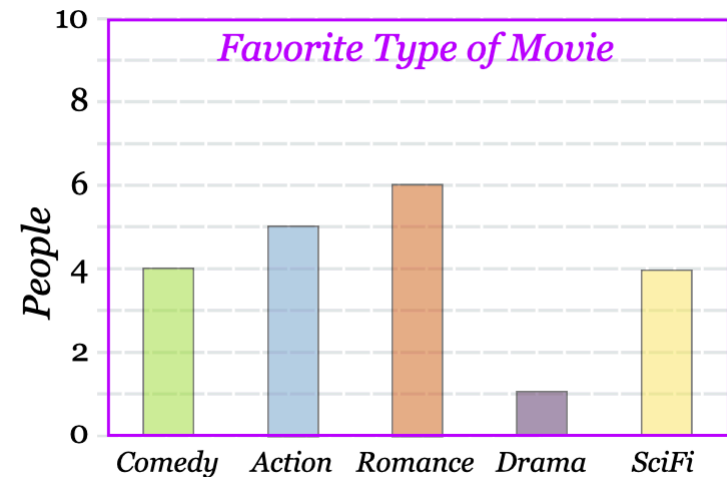
- Useful for comparing differences within groups
- x-axis shows categories, y-axis shows value scale (could be flipped)

Pie charts

- Useful for representing parts of a whole
 - percentages

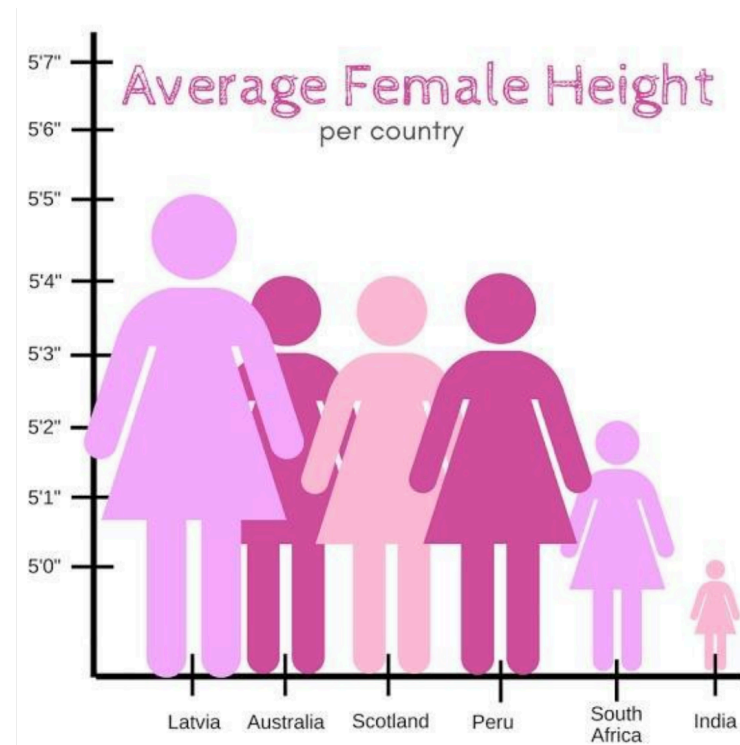
Bar plot

- Plots categorical values
 - Each category gets one bar
 - Length of the bar depicts “how many?”



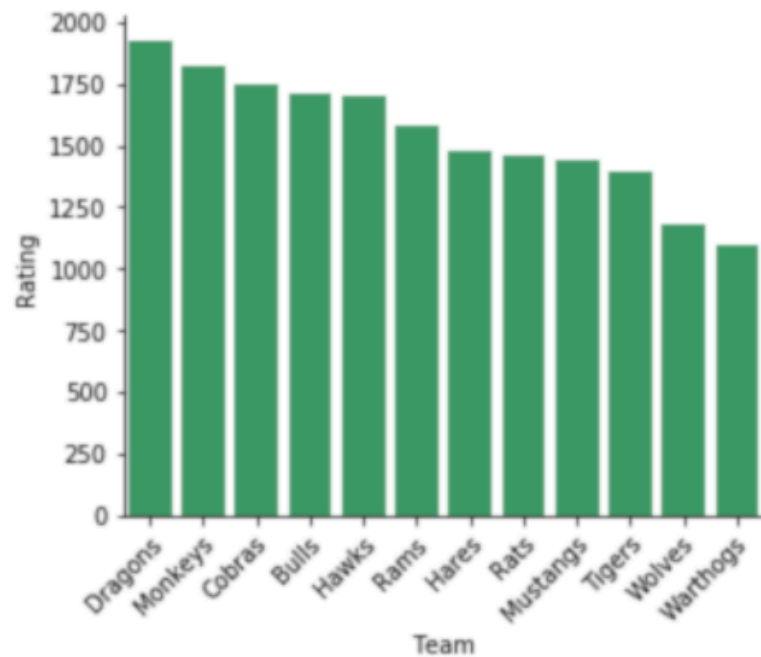
Be aware of possible issues

- Plotting against a nonzero baseline
- Replacing bars with images...



Horizontal or Vertical Bars?

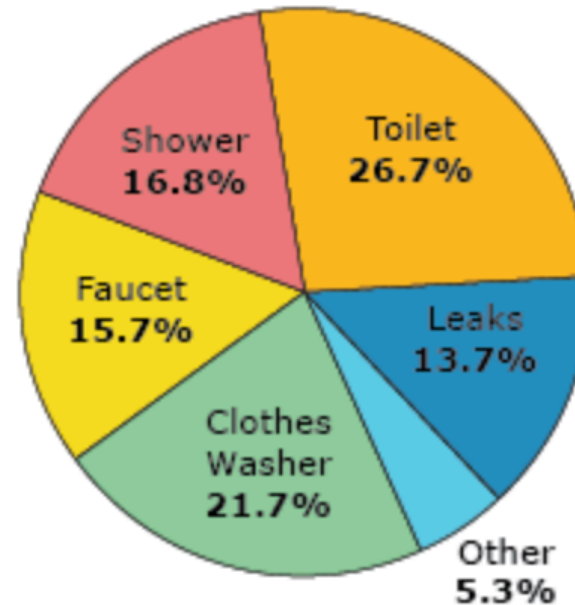
Useful for long category names



Pie chart

- showcases parts of a whole
- comparing groups to a whole (rather than group to group comparison)
- Good practice to include percentages

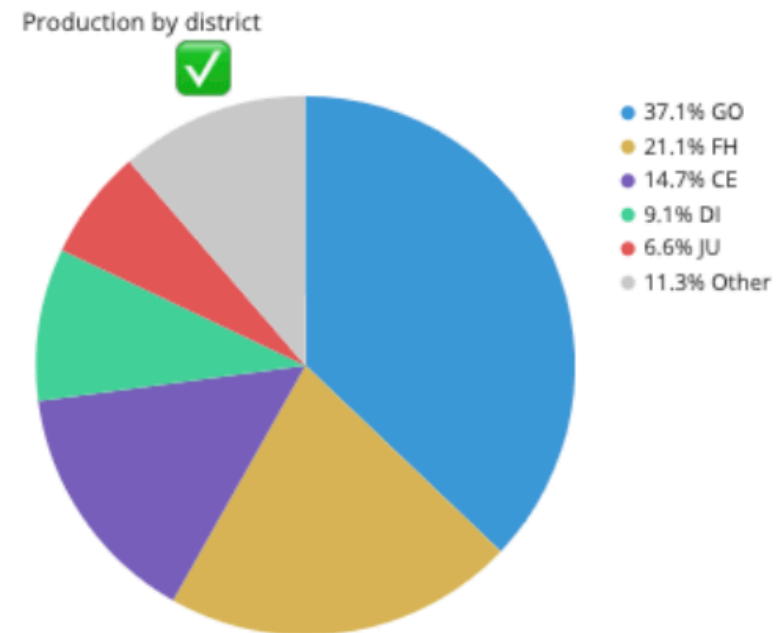
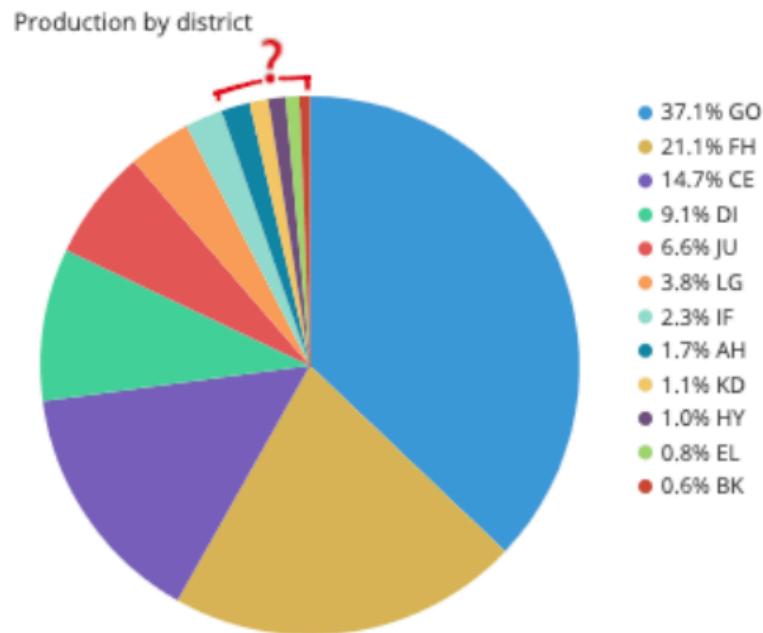
How Much Water Do We Use?



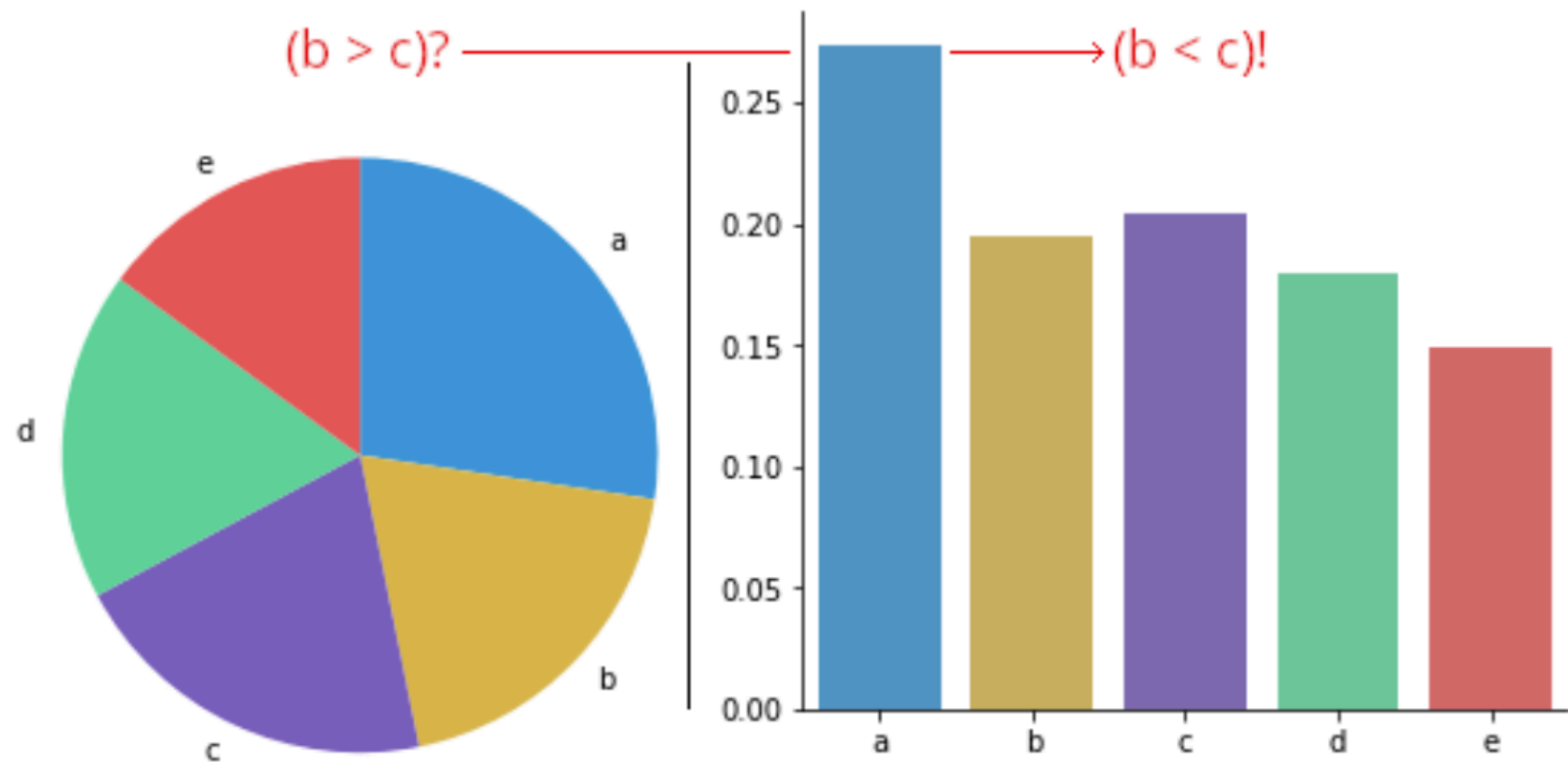
Source: American Water Works Association Research Foundation, "Residential End Uses of Water." 1999

Be aware of possible issues

- Large number of categories can make this difficult to read!

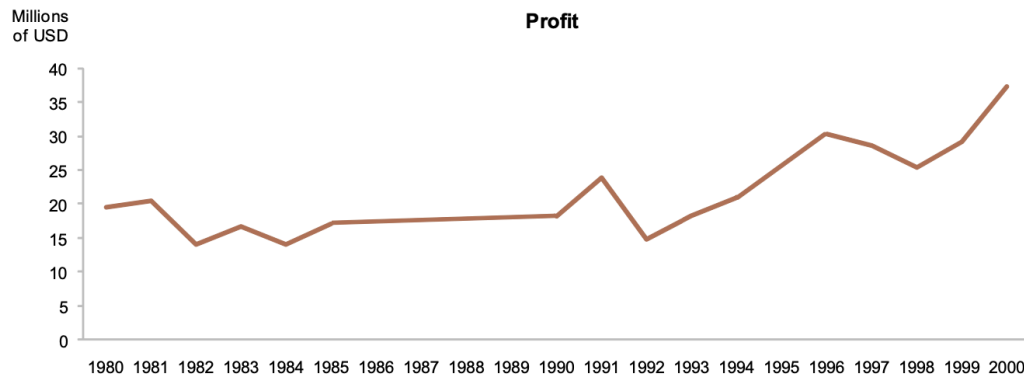
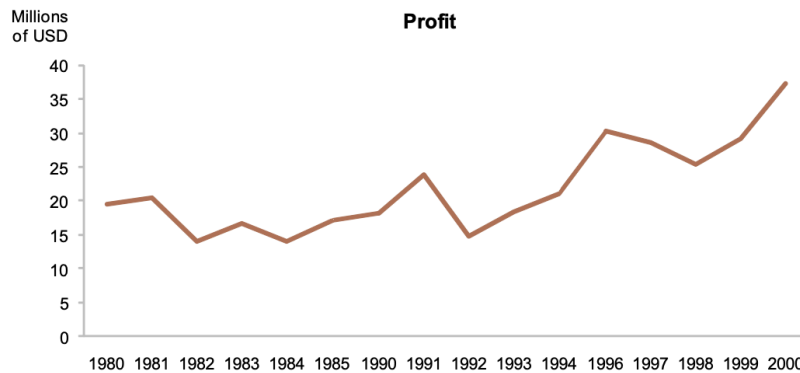


- Trying to compare between groups:



Be clear when you have missing data.

- Explain the reason why and any limitations.



In general

- Your graph highlights important conclusions!
- Include a brief description of the data along with your graph.

To see varying different types of graphs and a summary of when they might be useful, see:

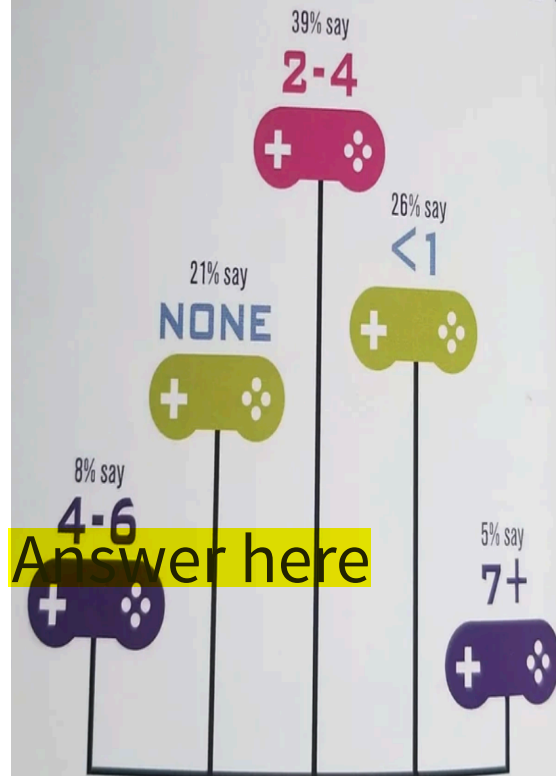
<https://datavizcatalogue.com>

Visualization Activity

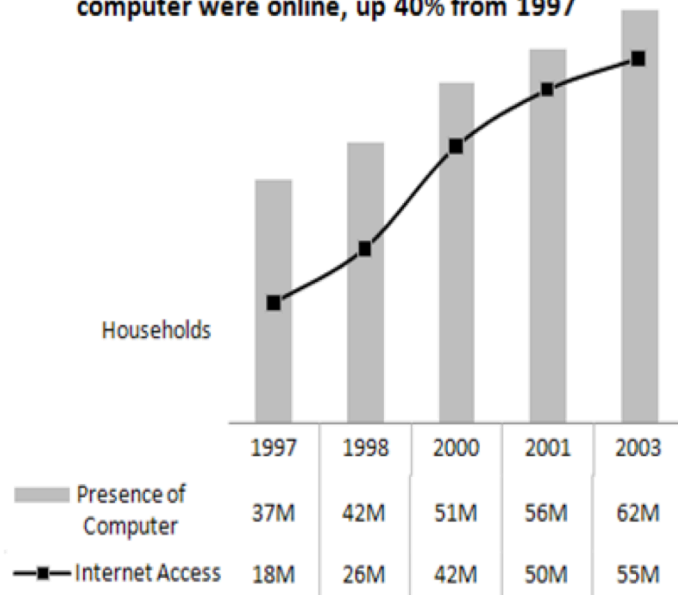
In the following graphs, discuss the questions below in groups:

1. What is wrong with the graph
2. How might you fix the graph to make the message more clear?

HOW MANY HOURS A DAY DO YOU SPEND GAMING?

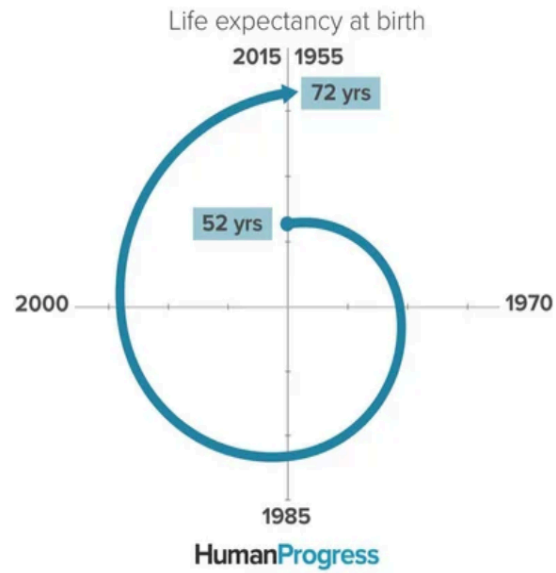


In 2003, more than 88% of households owning a computer were online, up 40% from 1997



Answer here

WE ARE **LIVING LONGER...**

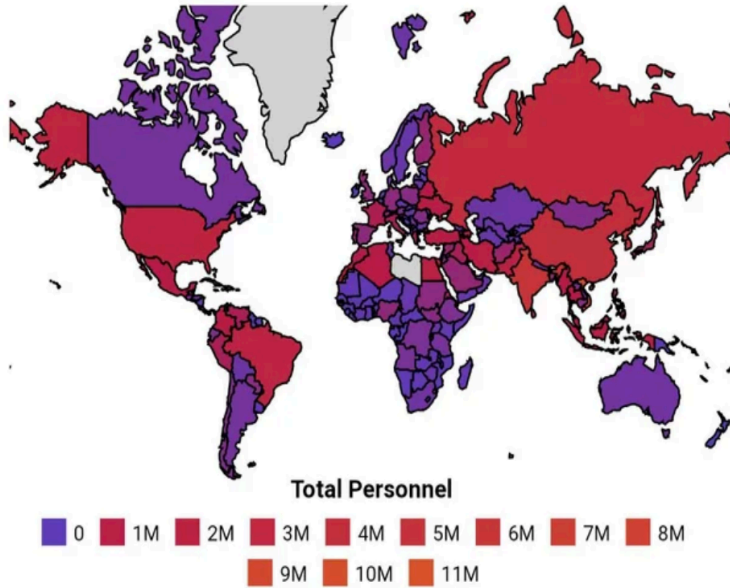


Answer here



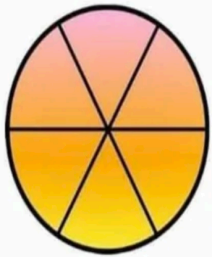
Answer here

Military Size by Country 2022



Answer here

THINGS THAT CHANGE OVER TIME



- Your appearance
- Your job title
- Your weight
- Your circle
- Your wealth
- Your size

THINGS THAT DON'T



- Your worth

Answer here